Tang Zhice

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EDUCATION

Chongqing University, Chongqing,China Sep. 2023 - Jun. 2027 (Expected)
Bachelor of Artificial Intelligence, School of Big Data & Software Engineering
GPA: 3.45/4.0, **Rank**: 7/53
Relevant Coursework: Deep Learning (95), Statistical Analysis (96), Data Structures (93), Programming Techniques (96)

RESEARCH INTERESTS

Computer Vision Multimodal Large Language Models Image and Video Generation
Embodied AI Medical Image Analysis

RESEARCH EXPERIENCE

Project: *"ASGP-SAM-Geo: Robust Interactive Medical Image Segmentation with Semantic Enhancement and Geometric Denoising"* ***Dec. 2024 - Present***

Project Leader

- Conducting research on **interactive abdominal multi-organ segmentation** based on SAM ViT-B for the **FLARE22** benchmark（**Larger datasets are being tested.**）, with a focus on improving the reliability of foundation models in medical imaging.
- Proposed a progressive framework addressing four key challenges in domain adaptation of foundation models: **cross-modal semantic alignment**, **robustness to noisy prompts, scale-aware fairness across organs**, and **smooth optimization under uncertain supervision**.
- Designed a **text-guided semantic enhancement mechanism** and an auxiliary prototype branch to inject anatomical prior knowledge into class-agnostic visual representations, enabling the model to move from generic object localization to semantic-aware organ segmentation.
- Developed a **Perturbation Module** and **Consistency Purifier** to simulate prompt noise and estimate interaction reliability, allowing the model to correct misleading prompts using intrinsic image features.
- Proposed an **AdaptiveSoftPurifier** with **scale-aware dynamic thresholds** and **soft weight routing,** transforming hard denoising decisions into continuous optimization for better robustness and training stability.
- Built a modular PyTorch training pipeline with per-sample loss routing, purity-driven prompt caching, and training-only auxiliary branches, introducing no additional inference cost.
- Achieved strong performance against **nnU-Net** on FLARE22, including higher **Dice** on the right kidney (+0.25), left kidney (+1.30), aorta (+6.72), and stomach (+1.96), as well as substantially higher **NSD** on the pancreas (+13.15), stomach (+13.91), aorta (+14.41), and esophagus (+8.84).
- A research manuscript based on this project is currently in preparation.

Project: "Real-time Face Recognition System" ***Aug. 2024 - Sep. 2024***

Project Developer

- Designed and implemented an end-to-end face recognition system using Python, TensorFlow/Keras, OpenCV, and Flask, covering data collection, model training, and web deployment.
- Built a real-time face data collection tool with OpenCV and Haar Cascade classifiers, enabling automated face detection, cropping, and per-person dataset organization.
- Constructed a two-layer CNN for multi-class face classification with max-pooling, a 1024-unit fully connected layer, Adam optimizer, and categorical cross-entropy loss.
- Deployed the trained model as a Flask-based web application supporting both image-upload

recognition and real-time video-stream recognition.

- Implemented preprocessing and system modules including grayscale normalization, adaptive interpolation, train-test split, exception handling, and resource management.

INDUSTRY EXPERIENCE

CITIC Securities, Beijing, China Jan. 2024 - Feb. 2024

Intern*(Remote) /*Full-time

- Assisted in industry research by collecting, cleaning, and organizing market and company data from multiple sources.
- Conducted data analysis and visualization to support sector trend tracking and investment research.
- Contributed to research reports by summarizing industry developments, comparing company performance, and presenting key findings in a structured and concise manner.

ACTIVITIES & AWARDS

The Second Prize Scholarship, Chongqing University Apr. 2024
The Third Prize Scholarship, Chongqing University Apr. 2025
Merit Student of Chongqing University Apr. 2025

Captain, National Flag Guard Team, Chongqing University
Sep. 2023 - Present

Led routine training, team coordination, and ceremonial activities, demonstrating leadership, discipline, and teamwork.
Supported major campus events requiring formal presentation and institutional representation.

LANGUAGES & SKILLS

Languages: Mandarin (Native), English（IELTS: 6.5）
Programming: Python, C++
Frameworks/Tools: PyTorch, TensorFlow/Keras, OpenCV, Flask, GitHub, Hugging Face
Other Skills: Data Analysis, Data Visualization, Microsoft Office (Word, Excel, PowerPoint)